

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY**Fifth Semester of B. Tech (EE) Examination****APRIL 2018****EE 305 Power Electronics and Drives - I****Date: 05.04.2018, Thursday****Time: 01:30 p.m. To 04.30 p.m.****Maximum Marks: 70****Instructions:**

1. The question paper comprises two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION – I

Q - 1 Write down the advantages and disadvantages of Power Electronics. **[07]**

Q – 2.a Explain different speed control methods for DC motor. **[04]**

Q – 2.b Answer any two questions. **[10]**

- (i) Draw and explain RC half-wave Triggering circuit for SCR
- (ii) Draw circuit symbols for the following power semiconductor devices.
BJT, MOSFET, Triac, IGBT, Thyristor
- (iii) Explain three-phase full-wave controlled rectifier with R load with all necessary waveforms and circuit diagram.

Q - 3 Answer any two. **[14]**

- a. Draw the schematics of step-up chopper and derive an expression for output voltage in terms of duty cycle.
- b. Draw circuit diagram for Single-phase half-wave thyristor controlled rectifier with R load. And find out rms output voltage equation.
- c. A step-up chopper has input voltage of 220 V and output voltage of 660 V. If the conducting time of thyristor-chopper is 100 μ s, compute the pulse width of output voltage.

SECTION – II

Q - 4 Draw and explain static I-V characteristics of a Thyristor with neat circuit diagram. **[07]**

Q – 5.a Explain working principle of step-down chopper with neat circuit diagram and waveforms. **[04]**

Q – 5.b Answer any two questions.

[10]

- (i) Explain External pulse Commutation of an SCR.
- (ii) Draw and explain concept of Electrical drive systems.
- (iii) Define Power Electronics and write down the Applications of Power Electronics.

Q – 6. Answer any two.

[14]

- a. Explain thyristor turn-on methods.
- b. Explain working principle of phase control in phase controlled rectifiers.
- c. Draw and explain triggering circuit for a Triac using Diac.
